American Computer Science League

2021-2022 ● Contest 1: Short Problems ● Intermediate Division

1. Computer Number Systems

Convert 2122_{10} to hexadecimal.

A. A48
B. B37
C. 849
D. 84A
E. 73A

2. Computer Number Systems

The Fibonacci sequence {0, 1, 1, 2, 3, 5, 8, 13, …} is formed by adding the two previous terms: 5 + 8 = 13. Which number in the sequence when converted to octal would have the 6th occurrence of the digit "1" when counting every digit?

A. 34
B. 55
C. 89
D. 144
E. 233

3. Recursive Functions

Find $f(20)$, given

$$f(x) = \begin{cases} 2 * f(x-3) + 1 & \text{if } x \geq 15 \\ \left[\frac{f(x+2)}{3}\right] & \text{if } 12 < x < 15 \\ 2 * x - 3 & \text{if } x \leq 12 \end{cases}$$

where [y] returns the greatest integer less than or equal to y

A. 43
B. 39
C. 19
D. 14
E. 9

4. Recursive Functions

Find $f(f(f(f(20))))$, given

$$f(x) = \begin{cases} f\left(\frac{x-2}{2}\right) - 1 & \text{if } x \text{ is even} \\ 2 * x & \text{if } x \text{ is odd} \end{cases}$$

A. 34
B. 18
C. 17
D. 12
E. 9

5. What DoesThis Program Do? (Branching)

What is printed when this program is executed?

```
a = 10 : b = 4 : c = 0 : d = 6
if a > b + d then
   b = b + d
else
   d = a - d
end if
if c ^ 2 < c ^ 4 then
   d = d * 2
else
   c = 1
end if
if b / d == c then
   c = c * 2
else
   a = a / c
end if
if b > c && d < b then
   d = d / 4
else
   a = c ^ 2 / d
end if
if d - b < a + b || b + c == d then
   d = d / 2
end if
x = a - b * c + d
output x
```

A. 7
B. 3
C. 1
D. -5
E. -9

美国计算机科学联赛

2021-2022 ● 第一轮：短题目 ● 中级组

1. 计算机数字系统

请将 2122_{10} 转化为十六进制。

A. A48
B. B37
C. 849
D. 84A
E. 73A

2. 计算机数字系统

斐波那契数列 {0, 1, 1, 2, 3, 5, 8, 13, ...} 由前两项相加而成，比如：5 + 8 =13。转化为八进制后，当每一位都算上时，该数列中第六次出现“ 1 ”的是哪个数？

A. 34
B. 55
C. 89
D. 144
E. 233

3. 递归函数

根据以下条件， 求 $f(20)$。

$$f(x) = \begin{cases} 2 * f(x-3) + 1 & x >= 15 \\ \left[\frac{f(x+2)}{3}\right] & 12 < x < 15 \\ 2 * x - 3 & x <= 12 \end{cases}$$

其中[y] 返回小于或等于 y 的最大整数。

A. 43
B. 39
C. 19
D. 14
E. 9

4. 递归函数

根据以下条件， 求 $f(f(f(f(20))))$。

$$f(x) = \begin{cases} f\left(\frac{x-2}{2}\right) - 1 & x\text{为偶数} \\ 2 * x & x\text{为奇数} \end{cases}$$

A. 34
B. 18
C. 17
D. 12
E. 9

5. 此程序是怎么运行的？（分支）

执行这个程序会打印输出什么？

```
a = 10 : b = 4 : c = 0 : d = 6
if a > b + d then
   b = b + d
else
   d = a - d
end if
if c ^ 2 < c ^ 4 then
   d = d * 2
else
   c = 1
end if
if b / d == c then
   c = c * 2
else
   a = a / c
end if
if b > c && d < b then
   d = d / 4
else
   a = c ^ 2 / d
end if
if d - b < a + b || b + c == d then
   d = d / 2
end if
x = a - b * c + d
output x
```

A. 7
B. 3
C. 1
D. -5
E. -9